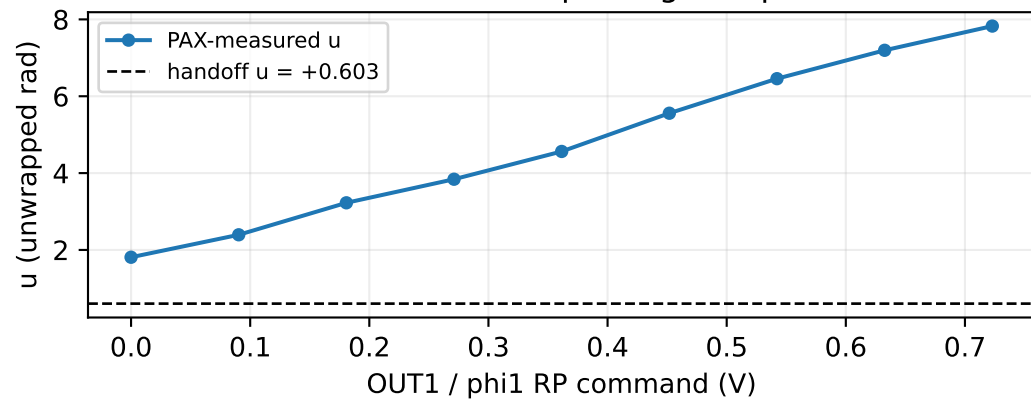
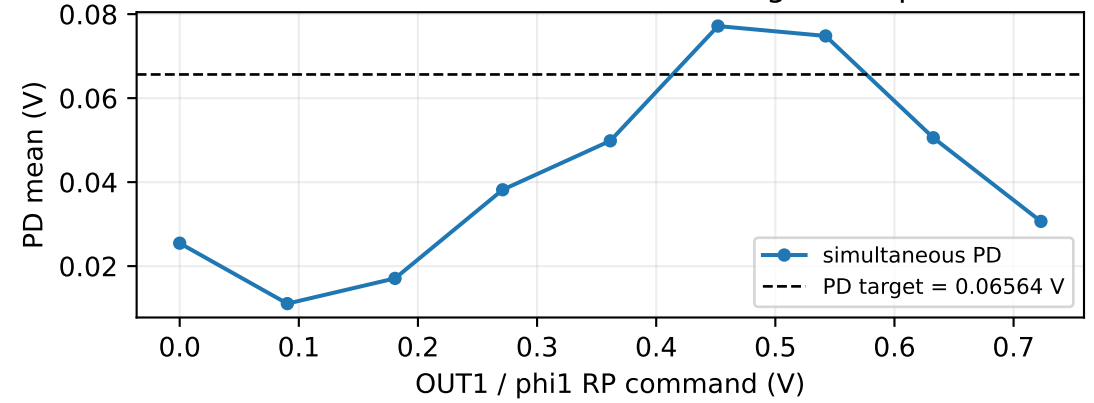


Phi1 PD FPGA lock — 012936_phi1-pd-lock-test_phi1-pd-lock-7

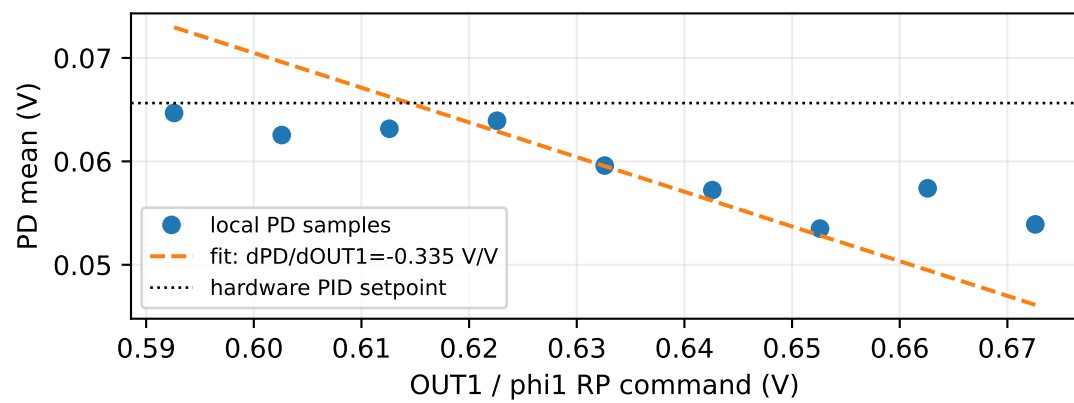
PAX-assisted ϕ_1 rough map



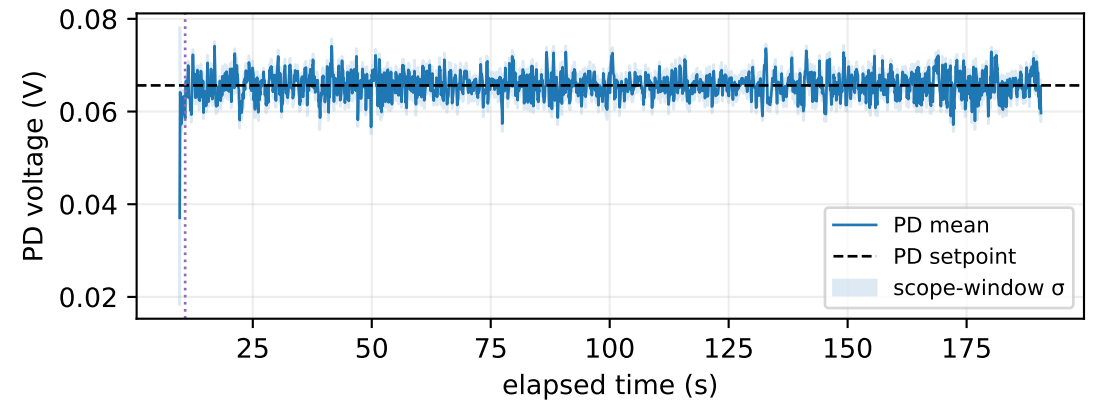
PD value associated with rough u map



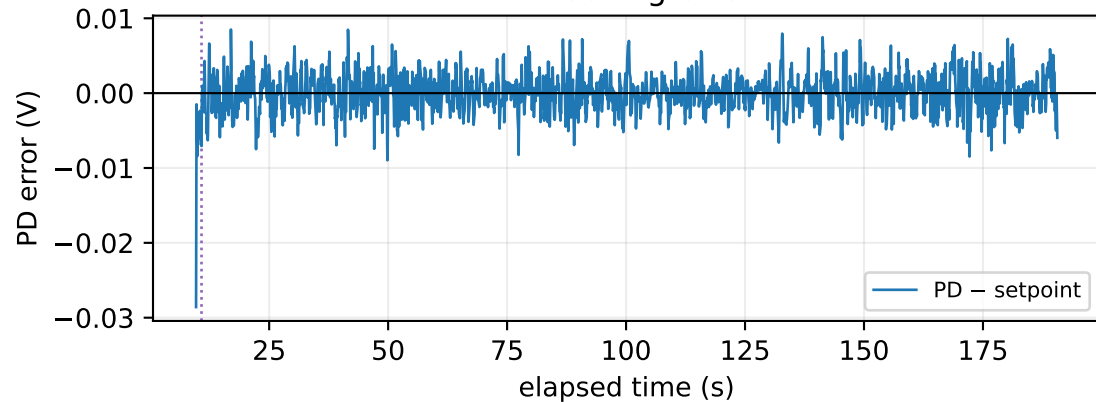
Automatic local PD calibration



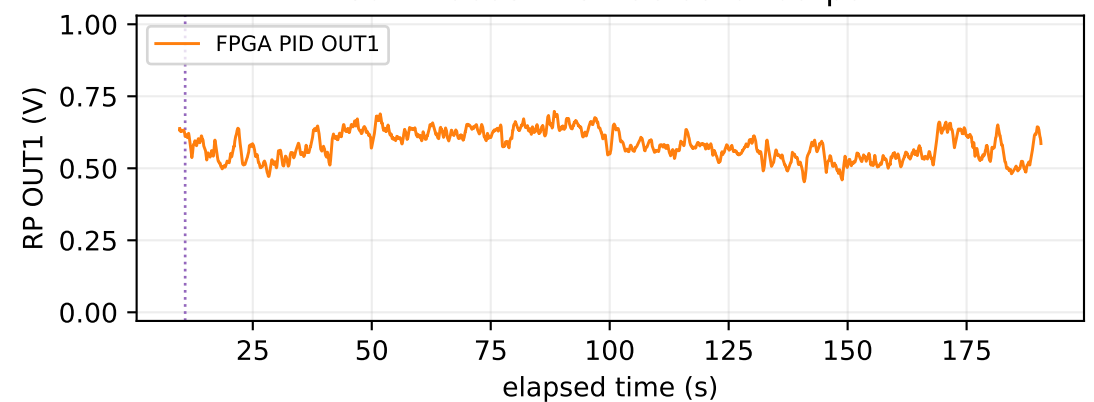
FPGA PD lock



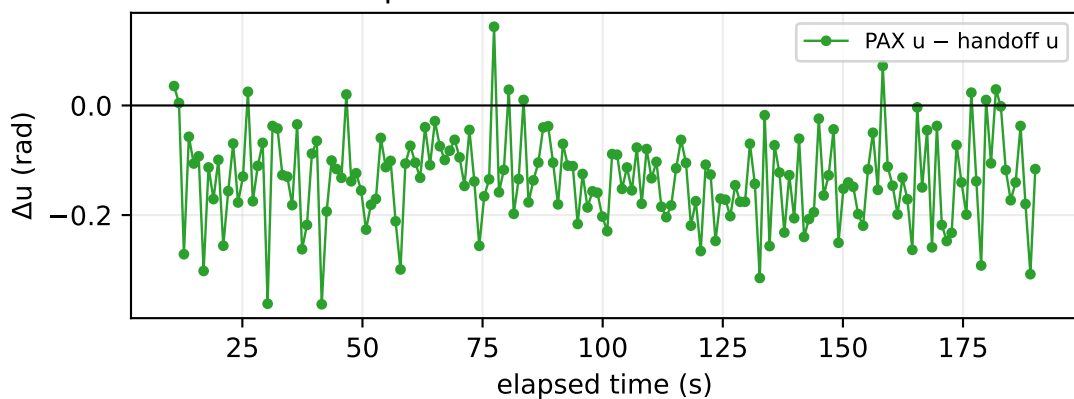
PD locking error



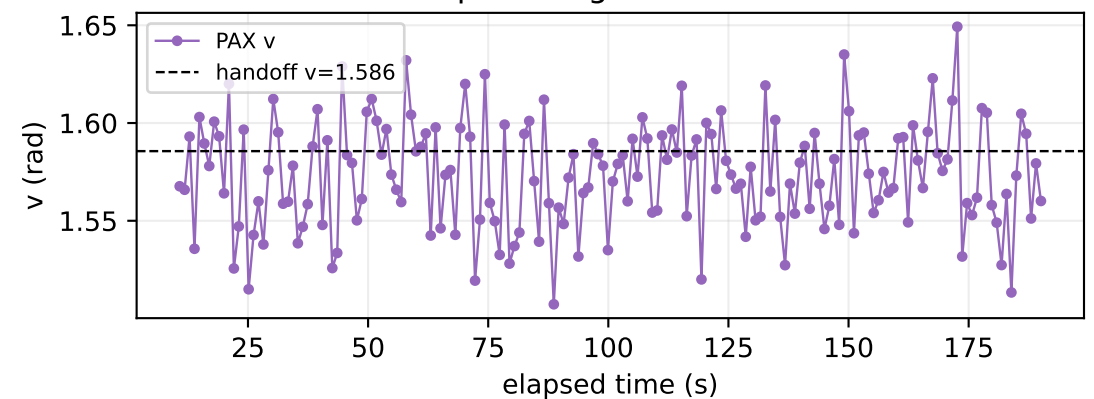
Continuous FPGA actuator output



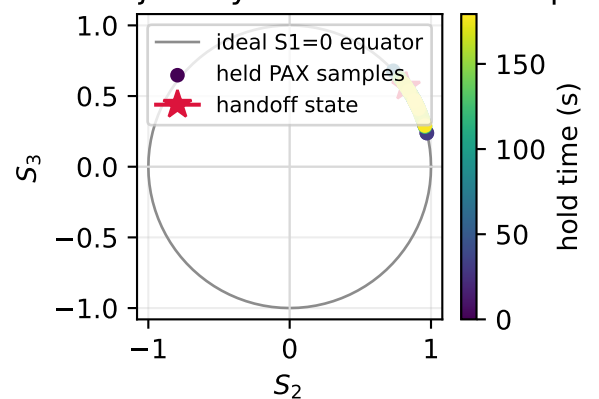
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



Interpretation

- Tight cluster near the red star: PD lock preserves the selected u branch.
- Motion around the circle: residual ϕ_1/u drift or branch movement.
- Radial motion: polarization is leaving the ideal equator.